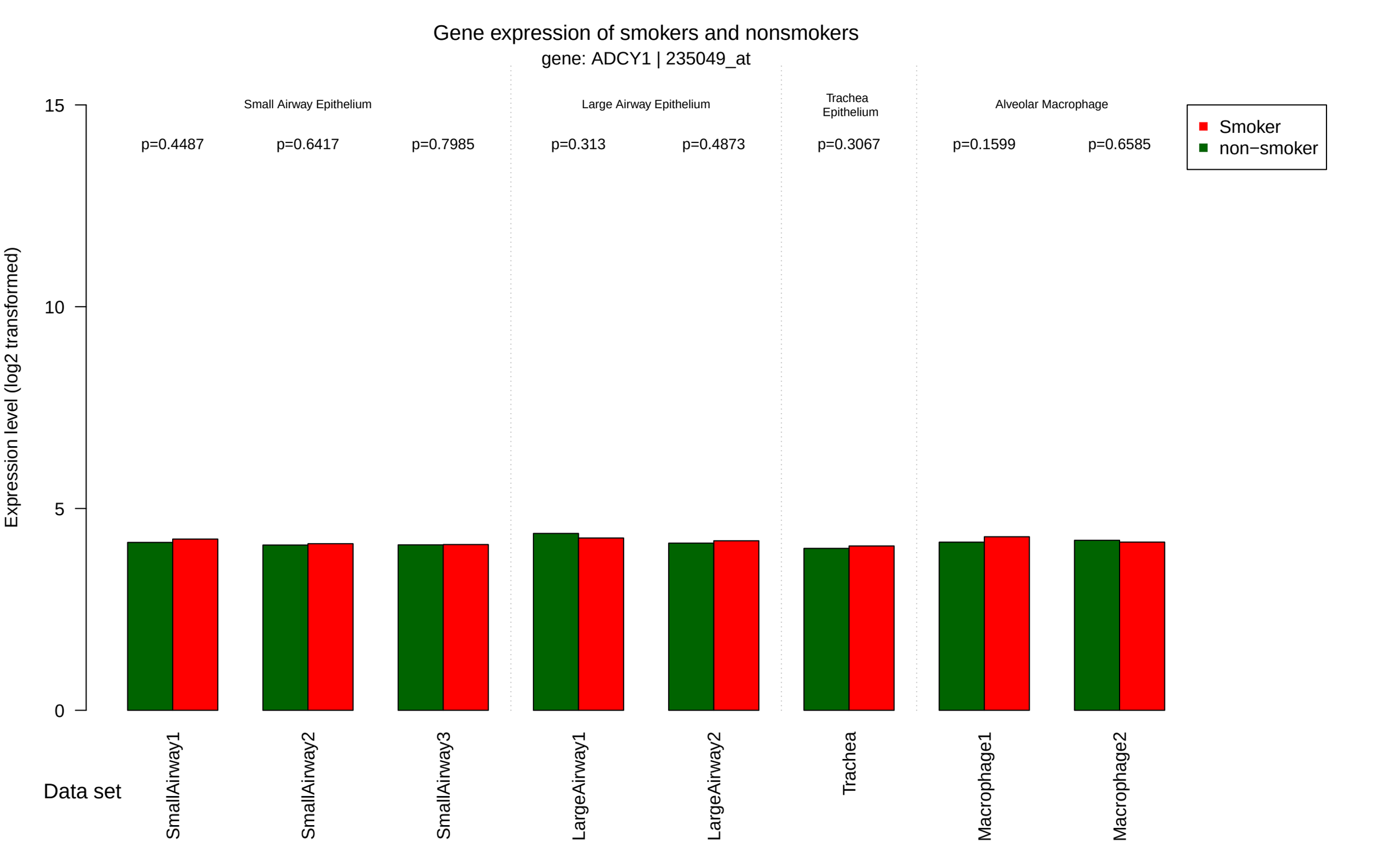
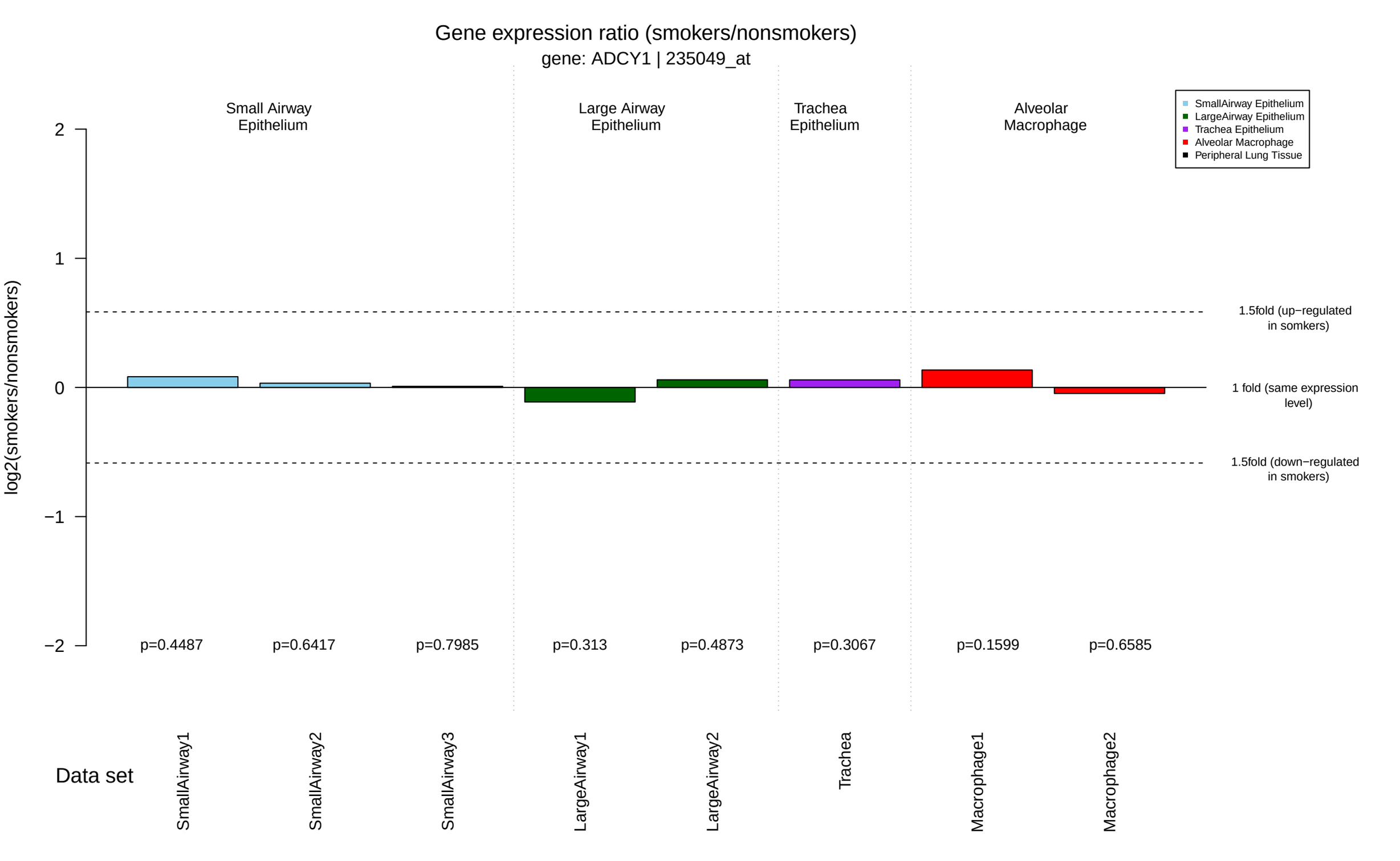
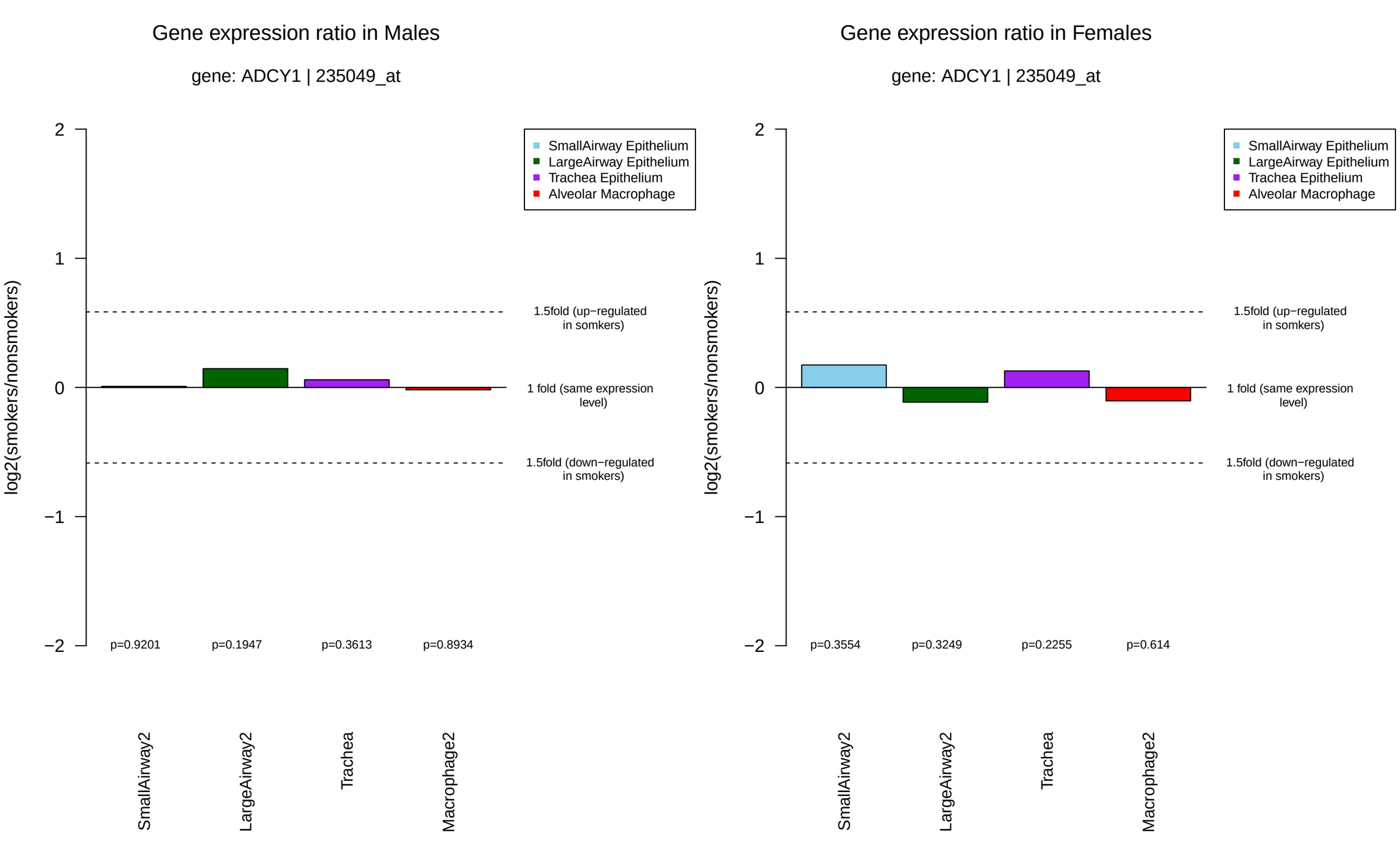
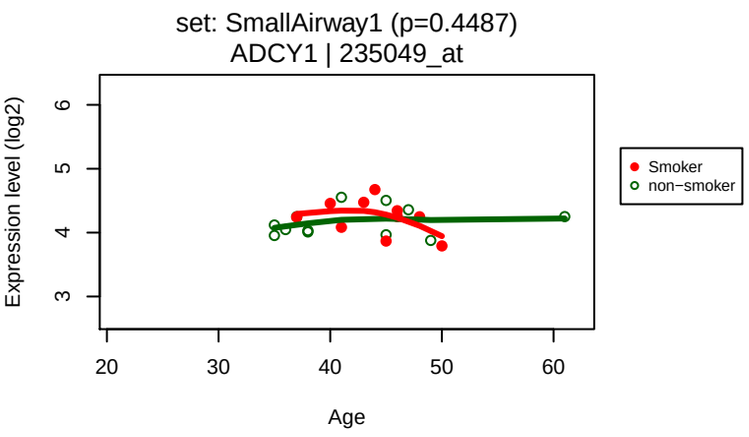
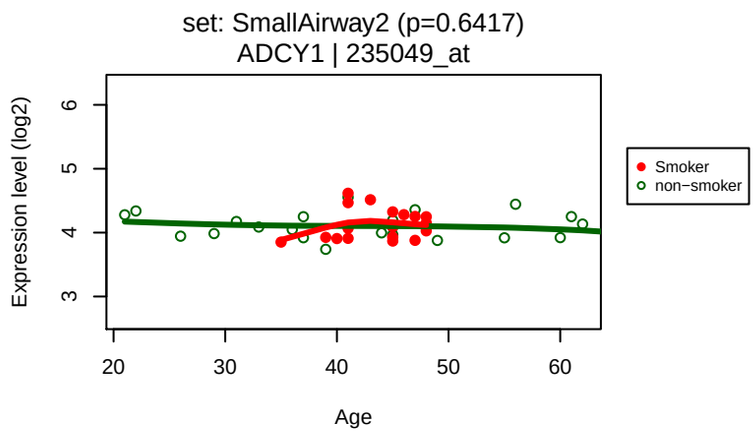
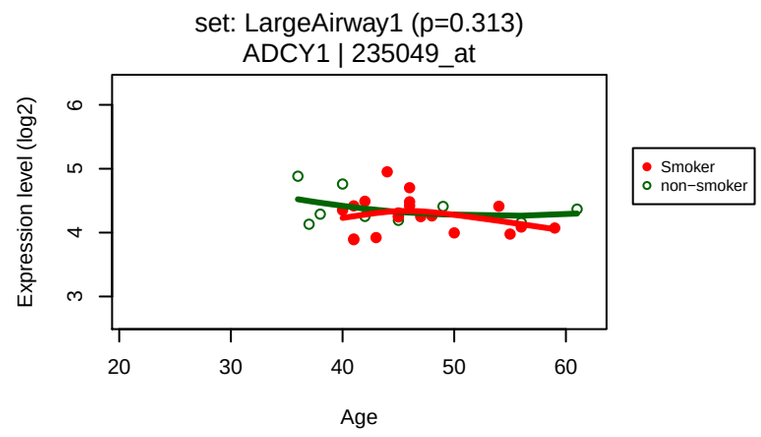

Gene expression of smokers and nonsmokers
gene: ADCY1 | 235049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4487
p=0.6417
p=0.7985
p=0.313
p=0.4873
p=0.3067
p=0.1599
p=0.6585
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

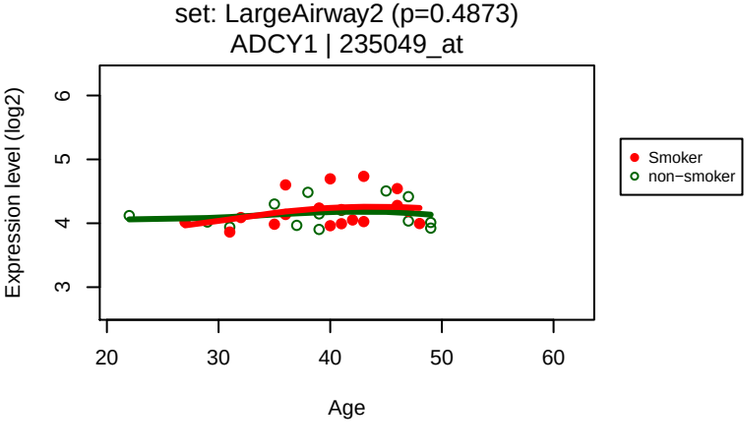

Gene expression ratio (smokers/nonsmokers)
gene: ADCY1 | 235049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4487
p=0.6417
p=0.7985
p=0.313
p=0.4873
p=0.3067
p=0.1599
p=0.6585
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

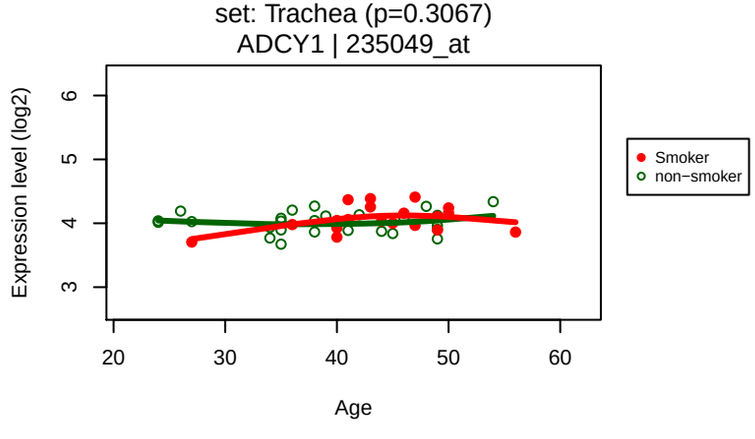

Gene expression ratio in Males
gene: ADCY1 | 235049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9201
p=0.1947
p=0.3613
p=0.8934
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY1 | 235049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3554
p=0.3249
p=0.2255
p=0.614
SmallAirway2
LargeAirway2
Trachea
Macrophage2

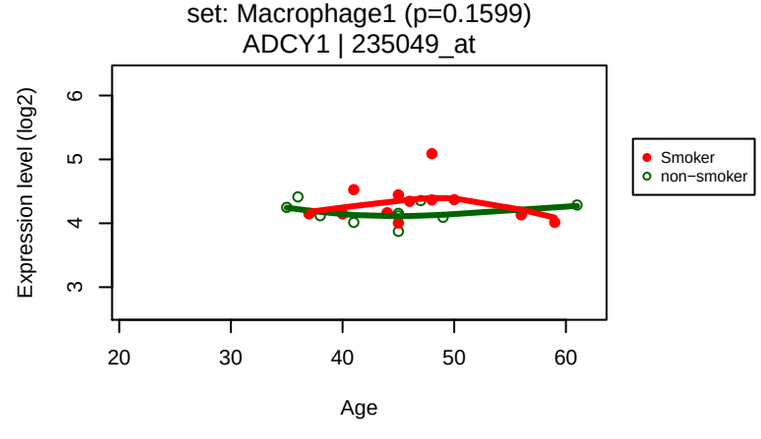

set: SmallAirway1 (p=0.4487)
ADCY1 | 235049_at
Expression level (log2)
Age
Smoker
non−smoker

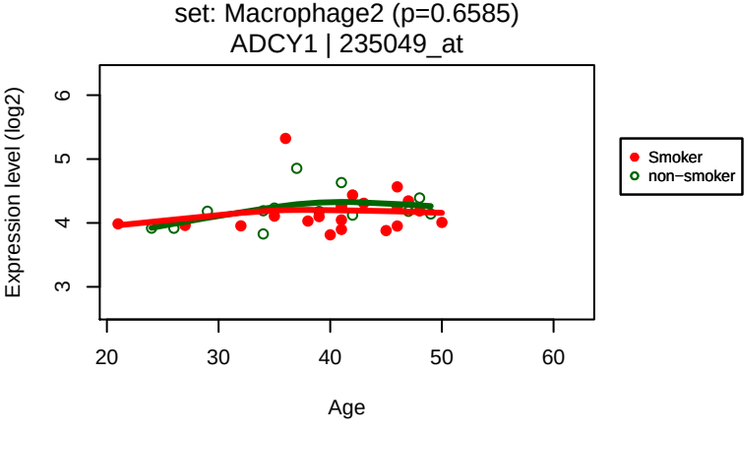

set: SmallAirway2 (p=0.6417)
ADCY1 | 235049_at
Expression level (log2)
Age
Smoker
non−smoker

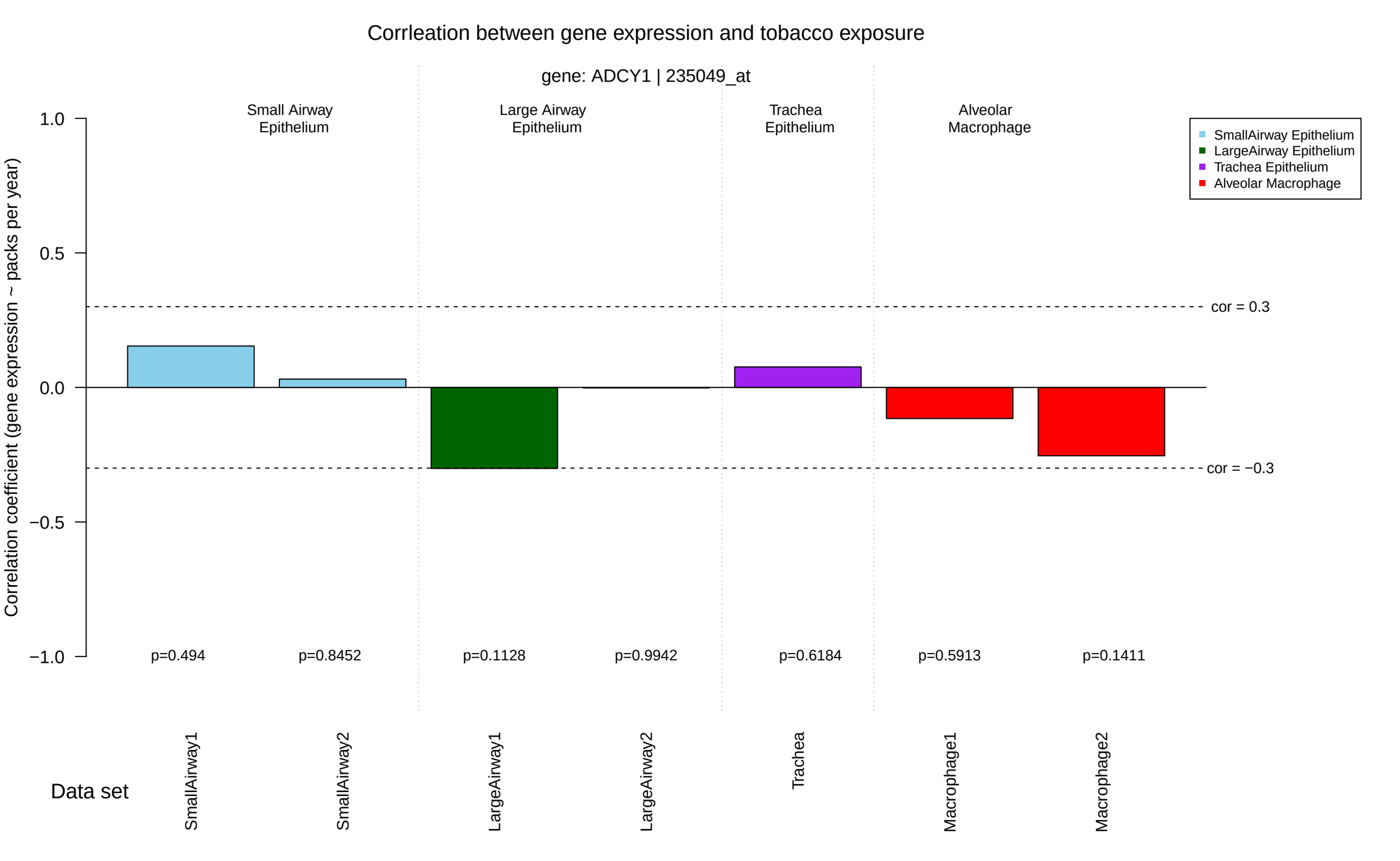

set: LargeAirway1 (p=0.313)
ADCY1 | 235049_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4873)
ADCY1 | 235049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3067)
ADCY1 | 235049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1599)
ADCY1 | 235049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6585)
ADCY1 | 235049_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY1 | 235049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.494
p=0.8452
p=0.1128
p=0.9942
p=0.6184
p=0.5913
p=0.1411
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2